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ADJUSTABLE GAS SYSTEM

Abstract

Disclosed is a firearm gas system including a gas block having a body with gas passageway for alignment with a barrel gas port. A gas tube has a gas inlet in a forward portion thereof and is axially variably positionable in at least partial alignment and disalignment relative to the gas passageway to thereby adjust gas flow.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/560,964, filed Mar. 4, 2024, and incorporates the same herein by reference.

TECHNICAL FIELD

[0002] This invention relates to an adjustable gas system for a gas-operated auto-loading firearm. In particular, it relates to such a system that is user-adjustable at the gas block and which changes the axial position of the gas tube.

BACKGROUND

[0003] Gas-operated firearms use a portion of the expanding propellant gas ported from the bore of the barrel through a gas block and a gas tube to cycle the action, either by direct impingement to a piston within the bolt carrier or by action of an external gas piston. Such systems are well known in the field.

[0004] For a given caliber of ammunition, the gas pressure and flow volume ported from the barrel through the gas block can vary significantly depending on the amount/type of propellant and weight of the projectile selected. Thus, it is at times desirable to be able to adjust the pressure and flow of gas ported from the barrel. Many others have provided flow adjustment mechanisms for a firearm gas operating system, including adjustments located in the gas block. Some of these mechanisms are complicated and costly to manufacture. Others provide only a limited number of discrete settings.

SUMMARY OF THE INVENTION

[0005] The present invention provides a system for adjusting the gas flow in a gas-operated self-loading firearm. The firearm gas system includes a gas block having a body with gas passageway for alignment with a barrel gas port and a gas tube having a gas inlet in a forward portion thereof. The gas tube is axially variably positionable in at least partial alignment and disalignment relative to the gas passageway to thereby adjust gas flow.

[0006] In a firearm, the system includes a barrel with an axial bore and a gas port that vents a portion of the propulsion gas pressure to a gas block and through a gas tube to the gas key of a bolt carrier. Gas flows through a gas passageway in the gas block and into an opening of the gas tube that is adjustably aligned with the passageway. The flow of gas is adjusted by longitudinal movement of the gas tube relative to the gas block, with operates to partially disalign the openings, thereby adjusting the cross-sectional size of the opening through which the gas can flow.

[0007] Other aspects, features, benefits, and advantages of the present invention will become apparent to a person of skill in the art from the detailed description of various embodiments with reference to the accompanying drawing figures, all of which comprise part of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Like reference numerals are used to indicate like parts throughout the various drawing figures, wherein:

[0009] FIG. 1 is a side view of a semiautomatic firearm (in this case, an AR-pattern rifle) including an embodiment of the present invention with the handguard shown in phantom for clarity;

[0010] FIG. 1A is a side elevation view of a semiautomatic firearm barrel and bolt carrier group with a gas tube and gas block according to one embodiment of the present invention;

[0011] FIG. 1B is an enlarged midportion view thereof;

[0012] FIG. 2 is an enlarged isometric view of the gas block and portion of the gas tube and barrel;

[0013] FIG. 3 is an exploded isometric view thereof;

[0014] FIG. 4 is a cross-sectional view taken substantially along line 4-4 of FIG. 1B;

[0015] FIG. 5 is a cross-sectional view taken substantially along line 5-5 of FIG. 1B;

[0016] FIG. 6 is a first fragmentary side sectional view of the gas block showing the gas tube in a first position;

[0017] FIG. 7 is a first fragmentary side sectional view of the bolt carrier group showing the gas

tube in a first position;

[0018] FIG. **8** is a second fragmentary side sectional view of the gas block showing the gas tube adjusted to a second position; and

[0019] FIG. **9** is a second fragmentary side sectional view of the bolt carrier group showing the gas tube adjusted to a second position;

DETAILED DESCRIPTION

[0020] With reference to the drawing figures, this section describes particular embodiments and their detailed construction and operation. Throughout the specification, reference to “one embodiment,” “an embodiment,” or “some embodiments” means that a particular described feature, structure, or characteristic may be included in at least one embodiment. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” or “in some embodiments” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the described features, structures, and characteristics may be combined in any suitable manner in one or more embodiments. In view of the disclosure herein, those skilled in the art will recognize that the various embodiments can be practiced without one or more of the specific details or with other methods, components, materials, or the like. In some instances, well-known structures, materials, or operations are not shown or not described in detail to avoid obscuring aspects of the embodiments. “Forward” will indicate the direction of the muzzle and the direction in which projectiles are fired, while “rearward” or “aft” will indicate the opposite direction. “Longitudinal” indicates a direction substantially parallel to the axis of the barrel along the length of the firearm. “Lateral” or “transverse” indicates a side-to-side direction generally perpendicular to the axis of the barrel. Although firearms may be used in any orientation, “left” and “right” will generally indicate the sides according to the user's orientation, “top” or “up” will be the upward direction when the firearm is gripped in the ordinary manner.

[0021] Referring first to FIGS. **1**, **1A**, and **1B**, therein is shown the gas system **10** of a typical AR-pattern firearm. The gas system **10** includes a barrel **12**, a gas block **14**, a gas tube **16**, and a bolt carrier group **18**. When the gas block body **26** is mounted on the barrel **12**, a gas passageway **28** is aligned with the barrel's gas port **22**. The illustrated embodiment is an internal piston (direct impingement) system in which gas pressure is delivered from the bore **20** of the barrel **12**, through a gas port **22**, through the gas block **14**, through the gas tube **16**, into the gas key **24** of the bolt carrier **18**, and into an internal cylinder where it acts against a piston head surface formed on the bolt to unlock the bolt and cycle the bolt carrier group to the rear in a well-known manner.

[0022] Referring now also to FIGS. **2-5**, the gas block **14** includes a body **26** with a primary bore **30** sized for mounting on a firearm barrel **12**. As is well known in the field, the gas block body **26** may be secured to the barrel **12** using threaded set screws **32**, for example, clamping (not shown), or other well-known means. Fluid communication with the barrel bore **20** is provided through a barrel port **22** aligned with a substantially radial gas passageway **28** in the gas block body **26**. The gas block body **26** includes a second longitudinal bore **34** that may be substantially parallel to the primary bore **30** for slidably receiving a forward portion of the gas tube **16**. In the illustrated embodiment, a gas tube **16** includes an inlet **36** in a sidewall and an end plug **38** with a cross bore **40** aligned with opposed openings **42** near the forward end of the gas tube **16**. The gas tube **16** is secured to a carrier block **44** with a pin **46** and, as described below, is longitudinally, selectably adjustable in position (along with the gas tube **16**) relative to the gas block body **26**. A pin **46** inserted through another cross bore **43** in carrier block **44** secures the gas tube **16** and end plug **38** to the carrier block **44** via the alignment of cross bores **40** and openings **42**.

[0023] The forward/aft position of the carrier block **44** and, thus, of the gas tube **16**, is adjusted by threaded engagement with an adjustment screw **48** situated inside an offset, threaded longitudinal opening **50**. The adjustment screw **48** is seated in a guide notch **58** and captured between forward and rear walls **52**, **54** of a niche **56** in the gas block body **26**. The adjustment screw **48** may be oriented substantially parallel to the secondary bore **34** and gas tube **16** and can be rotated in any

suitable manner, including manually. In the illustrated embodiment, the adjustment screw **48** can be rotated by a tool (not shown), such as a screwdriver or hex wrench. Because the adjustment screw **48** is substantially captive against longitudinal movement, rotation causes the carrier block **44** to move forward and aft, moving the gas tube **16** longitudinally with it. The secondary bore **34**, including in the front wall **52**, can act as a guide for the longitudinal sliding movement of the gas tube **16**. If desired, the screw **48** can have discrete rotational positions and held against unintended rotation by interaction of a detent **60** biased by a spring **62** (positioned in a blind end cross bore **64**) against longitudinal grooves **66** formed in the exterior threads of the adjustment screw **48**.

[0024] As previously described, gas pressure flows from the barrel bore **20**, through a radial barrel port **22** and the gas block passageway **28** to the gas tube inlet **36**. For full gas flow, the inlet **36** is fully aligned with the gas block passageway **28**, as illustrated in FIG. **6**. As the carrier block **44** and gas tube **16** are moved, the gas tube inlet **36** is be moved to or from alignment and disalignment with the gas block passageway **28**, changing the gas flow orifice size. As shown in FIG. **8**, when the gas tube **16** is shifted forward, the inlet is at least partially disaligned with the gas passageway **28**, reducing the gas flow orifice size and, thereby, reducing the gas flow.

[0025] As shown by a comparison of FIGS. **7** and **9**, the rear end position of the gas tube **16** that engages in and provides gas communication with the gas key **24** of the bolt carrier **18** also shifts. The standard, fully open gas flow position of the gas tube **16** is illustrated in FIGS. **6** and **7**.

Adjustment of the gas tube **16** to reduce gas flow moves it slightly forward, as shown in FIGS. **8** and **9**. The change in the rear end position of the gas tube **16** (at any adjustment setting) maintains engagement with the gas key **24** and the slight change in position has negligible (if any) effect on the operation of the gas system **10**.

[0026] While one or more embodiments of the present invention have been described in detail, it should be apparent that modifications and variations thereto are possible, all of which fall within the true spirit and scope of the invention. For example, the invention is not limited to an AR-pattern firearm nor to the gas tube being positioned above the barrel. Therefore, the foregoing is intended only to be illustrative of the principles of the invention. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not intended to limit the invention to the exact construction and operation shown and described. Accordingly, all suitable modifications and equivalents may be included and considered to fall within the scope of the invention, defined by the following claim or claims, interpreted according to the tenants of claim construction, including the doctrine of equivalents.

Claims

1. A firearm gas system, comprising: a gas block having a body with gas passageway for alignment with a barrel gas port; a gas tube having a gas inlet in a forward portion thereof, the gas tube being axially variably positionable in at least partial alignment and disalignment relative to the gas passageway to thereby adjust gas flow.
2. The gas system of claim 1, further comprising a threaded adjustment screw carried by the gas block.
3. The gas system of claim 2, further comprising a detent to retain the adjustment screw in discrete incremental rotational positions.
4. The gas system of claim 2, wherein the adjustment screw is axially offset from the gas tube.
5. The gas system of claim 1, wherein the gas tube inlet is located in a sidewall of the gas tube.
6. The gas system of claim 5, wherein the inlet is oriented downward.
7. The gas system of claim 1, further comprising a gas tube carrier block attached to a forward portion of the gas tube and configured to move the gas tube relative to the gas block body.
8. The gas system of claim 7, further comprising a threaded adjustment screw configured to adjust the position of the gas tube carrier block relative to the gas block body.

9. The gas system of claim 8, further comprising a detent to retain the adjustment screw at discrete rotational positions.
10. The gas system of claim 7, wherein the gas block body includes forward and rearward walls with a space between to receive the gas tube carrier block.
11. The gas system of claim 10, wherein gas tube acts as guide for the carrier block.
12. A semiautomatic, gas operated firearm, comprising: a receiver; a barrel having a longitudinal bore and a barrel gas port; a gas block having a body with gas passageway for alignment with a barrel gas port; a gas tube having a gas inlet in a forward portion thereof, the gas tube being axially variably positionable in at least partial alignment and disalignment relative to the gas passageway to thereby adjust gas flow.
13. The gas system of claim 12, further comprising a threaded adjustment screw carried by the gas block.
14. The gas system of claim 13, further comprising a detent to retain the adjustment screw in discrete incremental rotational positions.
15. The gas system of claim 13, wherein the adjustment screw is axially offset from the gas tube.
16. The gas system of claim 12, wherein the gas tube inlet is located in a sidewall of the gas tube.
17. The gas system of claim 16, wherein the inlet is oriented downward.
18. The gas system of claim 12, further comprising a gas tube carrier block attached to a forward portion of the gas tube and configured to move the gas tube relative to the gas block body.
19. The gas system of claim 18, further comprising a threaded adjustment screw configured to adjust the position of the gas tube carrier block relative to the gas block body.
20. The gas system of claim 19, further comprising a detent to retain the adjustment screw at discrete rotational positions.
21. The gas system of claim 7, wherein the gas block body includes forward and rearward walls with a space between to receive the gas tube carrier block.
22. The gas system of claim 21, wherein gas tube acts as guide for the carrier block.
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